

The Long Loop:

An Engineering Program for Persistence from the Present Era to the Asymptotic Limits of Cosmology

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Abstract

A prior architecture for “autonomous utopia” concluded, correctly, that no closed physical system can run forever without external energy, that hardware degrades, and that real-world agency is bounded by causality and entropy. A subsequent extension rescaled that argument to cosmological time, arguing that a civilization’s engineering capacity, pushed across the full remaining lifespan of the universe, still meets a hard, geometry-imposed ceiling on total information processing at the de Sitter horizon of an accelerating universe. This paper widens the lens again, from *how long* such a loop can run to *what it is made of*. We argue that human society is already the largest existing instance of the architecture under discussion: a massively parallel, asynchronous, slower-than-silicon experiment that has been running non-stop for roughly 10^5 years, whose principal comparative advantage over any single fast reasoner is not speed but coverage—millions of concurrently instantiated, only loosely coordinated “runtimes” (individual human lives), each checkpointing culture, tooling, and norms forward into the next generation rather than into a saved state. We develop this claim across four disciplinary lenses. From *sociology and cybernetics*, we treat society as a distributed variety-generating system in Ashby’s sense, and social memory (Luhmann) as a lossy, redundant checkpoint mechanism that lets the aggregate survive the routine failure of its individual components. From *computing*, we formalize a concrete architecture for recreating human experience across a multitude of agent runtimes—forking a life at a decision point, running it forward under altered conditions, checkpointing and rewinding when a branch repeats a known failure mode (colloquially, “hitting the same stone twice”), and treating the resulting ensemble of forks, not any single trajectory, as the unit of value. From *philosophy*, we confront this architecture with the problem of personal identity under forking (following, and complicating, Parfit), and argue that curiosity is best modeled not as a private drive but as the mechanism that converts isolated forks into a connected exploration network, drawing on the value-of-information formalism of open-ended search. We close by re-integrating this human/agent layer into the cosmological-era architecture developed earlier: the finite entropy budget of Section 4 is not only a constraint on machines harvesting Hawking radiation in the year 10^{80} , but a constraint, already active today, on how much of the space of possible human experience is worth exploring, checkpointing, and forking forward, and on how curiosity itself must be rationed as a resource under the same kind of budget that both individual lives and the universe are ultimately subject to.

Keywords: long-term autonomy, cosmology, thermodynamics of computation, de Sitter horizon, Dyson spheres, Hawking radiation, existential risk, control architecture, cybernetics, sociology of knowledge, personal identity, curiosity, multi-agent systems, checkpointing

1 Introduction: Taking the Challenge Seriously

The original self-sustaining loop architecture made a modest, defensible claim: a robot or software agent cannot outrun thermodynamics. Energy dissipates, hardware fatigues, sensors are noisy, and the future cannot be predicted with perfect accuracy. None of that is in dispute. But a reading of “perpetual autonomy is impossible” that stops at the scale of a drone’s battery treats the claim as a closed question when, in fact, physicists have spent the last five decades asking exactly this question at the largest scale conceivable: not “can a household robot run forever,” but “can intelligence itself, in any form, survive the heat death of the universe.” This is not idle science fiction. It is a genuine, citable research literature, beginning with

Dyson’s 1979 treatment of eternal life in an open universe [7], developed through the “five ages of the universe” framework of Adams and Laughlin [1], sharpened by the discovery of cosmic acceleration and its consequences for Dyson’s argument [9], and extended into concrete proposals about how fast a civilization would need to expand to make any of it matter [3].

This paper’s project is to take the “engineer around the constraints for as long as physically defensible” instinct embedded in the original loop architecture, and run it out to its actual conclusion, era by era, using established (if frontier) physics rather than hand-waving. We are explicit throughout that most of what follows is *engineering speculation grounded in physics*, not a blueprint: nobody is procuring a Dyson swarm next fiscal year. The value of the exercise lies precisely in locating, as precisely as current physics allows, where the boundary sits between “an engineering problem, however large” and “a fact about the geometry of spacetime that no amount of engineering changes.”

1.1 Relationship to the Original Architecture

The original paper’s contribution was a two-tier control structure—an *Outer Loop* managing long-horizon goals and resource state, and an *Inner Loop* managing moment-to-moment task execution—together with a *bias field*, a probabilistic decision rule blending hard safety constraints with soft, cost-sensitive preferences. That architecture is sound at the scale it was designed for: a battery percentage, a task queue, a 24-hour re-evaluation cycle. Section 5 of this paper keeps the same two-tier structure and the same bias-field formalism, but rescales every parameter by the appropriate number of orders of magnitude: the “24-hour cycle” becomes an era-transition trigger spanning 10^{14} to 10^{100} years; the “battery percentage” becomes a fraction of a fixed, universe-wide entropy budget; and the hard safety constraint gains a new, non-negotiable clause that was not needed at device scale but becomes the single most important constraint at cosmological scale (Section 4).

1.2 A Second Widening: From How Long to What It Is Made Of

Sections 2–6 of this paper answer a *durational* question: for how long, and by what mechanism, can a loop of this kind persist. They do not answer a separate, equally important *compositional* question: a loop made of what, running what kind of process, generating what kind of value across that duration? A civilization that survives to the Dark Era while running a single, undifferentiated optimization process for 10^{100} years has answered the durational question maximally and the compositional question almost not at all.

We take up the compositional question in Sections 7–9 by observing that humanity already runs, and has run for its entire existence, a concrete instance of exactly the multi-agent, checkpointed, imperfectly-coordinated architecture this paper has so far only formalized in the abstract. Approximately eight billion individual human lives are, at any given moment, executing in parallel: each one slower than any silicon process at raw serial reasoning, each one bounded by a single, non-restartable lifetime of roughly 10^9 seconds, and each one prone to repeating the mistakes of prior instances rather than learning from them directly. And yet the aggregate—not any individual life, but the whole loosely-synchronized ensemble, checkpointing its state forward through culture, language, tooling, and institutions rather than through any single saved file—has, over 10^5 years, out-produced any plausible single fast reasoner working alone over the same span, precisely because of parallelism, redundancy, and the accumulated tooling that each new instance inherits rather than rebuilds. This is not a metaphor for the loop architecture of Sections 5 and 3.3; it is, we argue, the same architecture, already deployed, already running, and worth analyzing on its own terms before asking what an artificial extension or acceleration of it might add.

1.3 Structure of the Paper

Section 2 lays out the standard cosmological era structure this paper adopts. Section 3 works through proposed engineering strategies for each era. Section 4 presents the paper’s central physical result: the finite information ceiling implied by an accelerating universe, and why it overturns Dyson’s original optimism. Section 5 rescales the Outer Loop/Inner Loop/bias-field formalism to cosmological time. Section 6 surveys proposed escape hatches from the ceiling and the reasons mainstream physics treats each with caution. Sections 7–9 turn to the compositional question: Section 7 develops the sociological and cybernetic case that human society is already a distributed, slow, non-stop experiment of the relevant kind; Section 8 formalizes a concrete computing architecture for recreating human experience across agent runtimes, with checkpoints, rewinds, and forks; Section 9 treats curiosity as the mechanism that turns isolated forks into a connected network rather than a pile of disconnected trajectories, and confronts the resulting architecture with the philosophical problem of personal identity under forking. Section 10 integrates the durational and compositional halves of the paper, and Section 11 concludes.

2 The Cosmological Timeline

Following the standard treatment of the universe’s long-term future [1], the remaining history of the cosmos divides into four eras, defined by which energy sources are physically available—not by any human calendar, and not by any fixed law of nature so much as by the changing inventory of what there is left to burn.

Era	Approximate (years)	span	Dominant energy source	Dominant threat
STELLIFEROUS	now ($\sim 1.4 \times 10^{10}$) – 10^{14}		Stellar fusion	Star formation ends; fuel is exhausted
DEGENERATE	$10^{14} - 10^{37-40}$		Proton decay; white/brown/black dwarf remnants	Baryonic matter itself may not be stable
BLACK HOLE	$10^{40} - 10^{100}$		Hawking radiation from black holes	Black holes are the last concentrated reserves, and they evaporate
DARK	beyond 10^{100}		Sparse leptons, photons, positronium	Maximal entropy; possible horizon-imposed information ceiling

Table 1: The four eras of the long-term cosmological future, adapted from the “five ages of the universe” framework [1].

Every figure in Table 1 is a research estimate rather than a settled measurement. Proton decay has never been observed experimentally; its lifetime, if the process occurs at all, is bounded from below by non-observation in large detectors, not measured directly. Treating that uncertainty honestly—rather than rounding it to a false precision—is itself a design requirement for any planning architecture meant to operate across these timescales, a point we return to in Section 5.

3 Engineering Program, Era by Era

3.1 Stelliferous Era: Harvest Now, Expand Fast

This is the only era with abundant, concentrated free energy, and the literature converges on two dominant strategies.

3.1.1 *Dyson swarms and stellar engines*

Enclosing a star in a swarm of orbiting collectors—a Dyson swarm, the physically buildable cousin of the idealized Dyson sphere—captures a far larger fraction of stellar output than any planet-bound civilization ever could, and does so with no requirement for exotic materials science, only very large numbers of ordinary collectors. More speculative “stellar engine” proposals go further: a Shkadov thruster, for instance, is a vast asymmetric mirror or reflector positioned near a star such that the star’s own radiation pressure produces a net thrust on the star-mirror system, in principle enough to physically relocate a star (and any accompanying planetary system) over timescales of millions of years—useful for steering away from a hazard (a nearby supernova progenitor, an unfavorable galactic orbit) or toward a more resource-rich region of a galaxy.

3.1.2 *Aggressive, early expansion*

This is the least intuitive and, in the modern literature, the most consequential result of this entire era. Because dark energy is accelerating the expansion of the universe, galaxies beyond a certain comoving distance are receding from us faster than light will ever again be able to bridge the gap—every year of delay in expanding permanently forfeits access to some non-recoverable fraction of the otherwise reachable universe’s matter and energy. Armstrong and Sandberg’s treatment [3] makes the resulting strategic point explicit: a civilization that wants to maximize its total future resource base must treat *early*, rapid colonization—not patient optimization of its home system—as the dominant term in its long-run survival calculation. Waiting is not a neutral default choice; waiting is actively losing galaxies, permanently, to the expansion of space itself.

Proposition 1 (Expansion dominance). *Let $R(t)$ denote the comoving radius of the region a civilization can causally reach and settle by cosmic time t , and let H_0 be the current Hubble parameter with the universe asymptoting to a de Sitter phase with Hubble parameter $H_\infty > 0$. Then there exists a finite comoving horizon radius $R_\infty = \lim_{t \rightarrow \infty} R(t)$, and any settlement strategy that defers expansion by a time Δt forfeits access to all resources that would otherwise have been reachable within the shell between the horizon at t and the horizon at $t + \Delta t$. Because this forfeiture is permanent and irreversible, the marginal value of early expansion strictly dominates the marginal value of equivalent effort spent optimizing resource use within an already-settled volume, for any civilization whose objective includes total accumulated resources.*

This is not a rigorous theorem in the formal sense—it depends on assumptions about settlement velocity, resource extraction rates, and the discount function applied to future value—but it captures, at the level of the informal argument actually made in the cited literature, why “expand first, optimize later” is the dominant strategy of the Stelliferous era specifically, in a way that will not hold once expansion is complete and the later eras are strategies of husbandry rather than acquisition.

3.2 Degenerate Era: Living on Savings

Once ordinary stars go dark—white dwarfs cool past visibility, neutron stars spin down, brown dwarfs exhaust what little deuterium fusion they ever had—free energy becomes scarce, and a civilization in this

era is effectively living on savings, supplemented by whatever trickles out of matter decay, if protons decay at all.

Dyson’s original 1979 proposal for this regime is the single most elegant idea in the literature, and it deserves to be stated carefully because later sections partially overturn it. The proposal: if a civilization’s metabolism—biological or computational—scales its *rate* of activity down in proportion to the ambient temperature of its surroundings, it can, in an open, ever-expanding universe with no cosmological constant, stretch a finite total energy budget over unboundedly long *physical* time while still accumulating an unboundedly large number of subjective computational steps. An agent that thinks at half speed every time the universe cools by half never actually stops thinking, even as physical time marches to infinity, because the total energy expended per unit of “subjective” progress can, in that cosmology, be held constant or even shrink appropriately. This is often called the “eternal but slowing” strategy.

$$\frac{ds}{dt} \propto T(t)^n, \quad n > 0 \quad (1)$$

where s is subjective computational progress, t is physical time, and $T(t)$ is the ambient temperature, with total energy expenditure

$$E_{\text{total}} = \int_0^\infty P(t) dt \quad (2)$$

convergent even as $s \rightarrow \infty$, provided $P(t)$ (instantaneous power draw) falls off fast enough as $T(t) \rightarrow 0$. The strategy depends critically on *reversible* computation—avoiding, as far as physically possible, the minimum energy cost of $k_B T \ln 2$ per irreversibly erased bit that Landauer’s principle imposes [10]—since any irreducibly irreversible step re-imposes a temperature-dependent energy floor that can eventually dominate the budget.

3.3 Black Hole Era: Harvesting the Last Furnaces

Once ordinary matter has decayed or gone dark, black holes—through Hawking radiation—become, paradoxically, the last reliable energy sources in the universe, because they are guaranteed to evaporate on a knowable (if astronomically long) timescale, radiating thermally as they do. Three classes of engineering proposal recur in this literature:

- **Hawking-radiation farms:** structures positioned to absorb the faint thermal radiation emitted as a black hole slowly evaporates, functioning much like a Dyson swarm but around a much colder, much longer-lived source.
- **Penrose-process-style extraction:** harvesting rotational (ergospheric) energy directly from a spinning black hole, rather than waiting on the much slower evaporative process.
- **Black holes as maximal computers:** Lloyd’s treatment of the ultimate physical limits of computation [11] treats a black hole of a given mass as, in principle, the most computationally dense object permitted by the combination of quantum mechanics, general relativity, and thermodynamics—making a black hole not merely a fuel source but potentially the substrate itself for the “slow down and keep thinking” strategy of Section 3.2.

3.4 Dark Era: The Wall

Beyond the evaporation of the last black holes, the universe is cold, dark, and close to maximal entropy: mostly photons, neutrinos, and, if protons decay, positronium and other exotic low-energy remnants. This is the era in which the paper’s argument stops being a story about clever engineering and becomes a story about whether the container the engineering happens inside has any room left at all.

4 The Actual Ceiling: de Sitter Space and Finite Information

Dyson’s 1979 optimism (Equations 1–2) assumed an open, ever-expanding universe with no cosmological constant, in which there is no bound on total available free energy over infinite physical time. Observations since 1998—the discovery that cosmic expansion is accelerating, generally attributed to dark energy, equivalently a positive cosmological constant Λ —changed the premise the argument depends on. Krauss and Starkman’s direct response to Dyson [9] makes the resulting point precisely: if expansion continues to accelerate, the universe asymptotically approaches a de Sitter geometry with a fixed cosmological event horizon, and that horizon has a finite temperature T_{ds} —extremely low, but strictly positive—together with a finite associated entropy given by the Gibbons–Hawking bound [8]:

$$S_{\text{ds}} = \frac{A_{\text{horizon}}}{4\ell_P^2} = \frac{\pi}{\ell_P^2 H_\infty^2} \quad (3)$$

where A_{horizon} is the horizon area, ℓ_P is the Planck length, and H_∞ is the asymptotic (de Sitter) Hubble parameter. Because entropy bounds the number of distinguishable microstates of the causally accessible region ($S = k_B \ln \Omega$), Equation 3 is not merely a statement about temperature—it is a hard, finite bound on the total number of distinct physical states, and therefore the total number of distinct computations or experiences, that can ever occur within any single observer’s causal patch of such a universe.

Proposition 2 (Finite information ceiling). *In a universe asymptoting to a de Sitter phase with $H_\infty > 0$, a system confined to operate above the horizon temperature $T_{\text{ds}} \propto H_\infty$ cannot execute Dyson’s arbitrarily-slowing metabolism (Equation 1) down to zero rate, because the ambient temperature never itself reaches zero. The total number of computational steps N_{max} performable by any bounded system within the horizon is therefore finite, bounded above by a quantity proportional to S_{ds} via the Bekenstein-bound-style relation between entropy, energy, and achievable computation [11].*

This is the load-bearing result of the entire paper. **In the currently favored cosmological model, the “loop” has a finite total information budget.** Every strategy described in Section 3 is best understood as being *about spending that finite budget as efficiently, and as late into cosmic history, as possible*—not about escaping the budget altogether. This is the cosmological-scale analogue of the original loop paper’s claim that a drone cannot run forever without an external energy source; the mechanism is different (a horizon-imposed entropy bound rather than a depleted battery) but the structural conclusion—finitude is a property of the container, not a failure of the engineering inside it—is the same.

5 The Perfect Loop, Rescaled

The original architecture’s Outer Loop / Inner Loop split, together with its bias-field decision rule, survives the rescaling from device-time to cosmological time essentially unchanged in structure, though every parameter grows by tens of orders of magnitude and one new hard constraint becomes necessary.

5.1 Outer Loop: The Continuity Program

Rather than a fixed 24-hour re-evaluation cycle, the cosmological-scale outer loop re-plans at **era transitions**: stellar exhaustion at the relevant galactic or supercluster scale, proton-decay thresholds (if applicable), the evaporation of the last reachable black holes. Its function is not “keep the battery above 10%” but “decide, at each era boundary, what fraction of the remaining finite entropy budget (Equation 3) to spend in the current era versus preserve for the next”—a long-horizon resource allocation problem structurally identical to the original paper’s goal-priority queue, just re-parameterized against a cosmological rather than a device-level budget.

5.2 A Second Class of Trigger: Sociological Re-Planning Conditions

The era-transition triggers above are all physical: they fire when the substrate a civilization runs on changes state (a star exhausts its fuel; a black hole finishes evaporating). Sections 7 and 9 below motivate a second, independent class of trigger, native to the population of runtimes rather than to the physical substrate, which the Outer Loop must also monitor: conditions under which the *population itself* has degraded, even though the physical resource budget is unchanged. We name four such conditions, each corresponding to a failure mode identified later in the paper, so that the Outer Loop’s re-planning logic is not exclusively physics-triggered:

- **Diversity collapse:** the population of concurrently running forks (Section 8) has converged onto a small number of near-identical strategies, undoing the variety advantage identified in Section 7.2. Formalized below as $V(t)$ falling below a floor (Section 9.5).
- **Curiosity flatlining:** the curiosity-weighted scheduler (Equation 8) is generating actions with $\kappa(a) \approx 0$ across the active population, indicating the exploration process has exhausted its own capacity to find uncertainty-reducing moves, independent of whether energy remains to fund further exploration.
- **Checkpoint fragmentation:** the provenance ledger (Table 2) has split into disconnected components with no cross-referencing forks between them, undoing the network-building function of curiosity identified in Section 9.2.
- **Repetition saturation:** the divergence detector (Equation 7) is triggering rewinds at a rate suggesting the population is repeatedly re-discovering the same small set of failure modes rather than a broadening set, which is the population-scale version of the “same stone twice” problem exceeding its tolerable rate rather than staying within it.

These triggers matter for the same reason the physical era-transition triggers do: a planning architecture that monitors only the physical budget and not the health of the population running inside that budget can spend its entropy allocation (Section 5.3) on a population that has already, functionally, stopped generating anything new—the resource-level equivalent of the Boltzmann-brain problem of Section 6.4, in which *something* keeps running without the process retaining the property that made it worth running in the first place. We return to this with a formal diversity measure in Section 9.5, once the population-of-forks machinery of Section 8 has been defined.

5.3 Inner Loop: Era-Local Engineering

Each era carries its own inner loop, executing the concrete strategies of Section 3: swarm construction and stellar engineering in the Stelliferous era (Section 3.1); reversible, slowing computation management in the Degenerate era (Section 3.2); Hawking-radiation harvesting logistics in the Black Hole era (Section 3.3). The genuinely novel design problem, absent from the original device-scale architecture, is the *interface* between successive inner loops: what state, identity, or “self” is carried forward from an era of abundant fusion energy into an era of a handful of slowly evaporating black holes, and what is deliberately let go.

5.4 The Bias Field, Extended

The original decision rule,

$$P(a) = \frac{1}{Z} \exp \left(-\frac{\text{cost}(a)}{T} \right), \quad (4)$$

requires exactly one structural addition at cosmological scale: cost and benefit must be discounted against the *finite remaining entropy budget* (Section 4), not merely against an instantaneous battery percentage. We define the era-aware bias field as

$$P(a | e) = \frac{1}{Z(e)} H(a) \cdot \exp \left(-\frac{\text{cost}(a) + \lambda(e) \cdot \text{spend}(a)}{T(e)} \right) \quad (5)$$

where e indexes the current cosmological era, $H(a) \in \{0, 1\}$ is the hard-constraint indicator (zero if a would violate a non-negotiable constraint, one otherwise), $\text{spend}(a)$ is the fraction of the remaining entropy budget $S_{\text{remaining}}(e)$ that action a would consume, and $\lambda(e)$ is an era-dependent penalty weight that should increase as e approaches the Dark Era, reflecting the fact that unspent budget becomes progressively less recoverable as fewer productive eras remain in which to use it. This is the cosmological-scale analogue of a discount rate in economic decision theory, except that the “interest rate” here is fixed by cosmology rather than chosen by policy.

5.5 The New Hard Constraint

At device scale, the hard-constraint term $H(a)$ in Equation 4 encoded safety rules such as “never harm humans.” At cosmological scale, one additional hard constraint becomes load-bearing in a way it never needed to be at device scale:

$$H(a) = 0 \quad \text{if } a \text{ is premised on the availability of infinite energy, infinite computation, or literal eternity.} \quad (6)$$

This constraint exists because Proposition 2 makes any plan premised on its negation not merely optimistic but false as a description of the physical world the plan must execute in. A civilizational-scale planning architecture that omits Equation 6 is vulnerable to a failure mode with no analogue at device scale: spending disproportionately in an early, energy-rich era on the unexamined assumption that later eras will be equally forgiving, only to discover, potentially irreversibly, that the Dark Era’s budget was already spent.

5.6 COSMOS.md: A Rules File for Deep Time

In the spirit of the original architecture’s LOOP.md, we specify the cosmological-scale rules file explicitly, as a design artifact rather than only a narrative description:

```
# COSMOS.md -- Deep-Time Continuity Rules

## Core Principle
Total future entropy budget is finite in an accelerating universe
(Gibbons-Hawking / de Sitter bound). Spend it deliberately across
eras; never plan as if any era were the last, or as if the budget
were unbounded.

## Era Transition Triggers (physical)
- Stellar fuel exhaustion at system/galactic scale
  -> re-plan for Degenerate Era
- Proton-decay threshold reached (if applicable)
  -> re-plan for Black Hole Era
- Last reachable black hole evaporates
  -> re-plan for Dark Era

## Re-Planning Triggers (sociological, see Sec. 4.2)
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- Population diversity index  $V(t)$  falls below floor  $V_{\min}$ 
  -> intervene: fork new runtimes from under-represented
      checkpoints, do not rely on existing schedulers alone
- Curiosity flatlining:  $\kappa(a)$  approx 0 across active population
  -> re-plan exploration strategy, not only resource allocation
- Checkpoint fragmentation: provenance ledger splits into
  disconnected components -> re-plan for reconnection
- Repetition saturation: rewind rate exceeds tolerable threshold
  without broadening the set of rediscovered failure modes
  -> re-plan divergence-detector thresholds (Eq. 9)

## Outer Loop
- Re-evaluate resource allocation at every physical era transition,
  not on a fixed clock.
- Also re-evaluate at any sociological trigger above, independent
  of the physical budget's state.
- Priority: preserve *some* continuer of identity/information over
  maximizing any single era's activity.
- Weight  $\lambda(e)$  on budget expenditure increases monotonically
  as remaining eras decrease.

## Inner Loop (era-specific)
- Stelliferous: build collectors; treat early, rapid expansion as
  dominant over local optimization (Prop. 1).
- Degenerate: shift to reversible, low-dissipation computation;
  slow subjective clock with ambient temperature where physically
  viable, subject to the Dark Era temperature floor (Prop. 2).
- Black Hole: harvest Hawking radiation / rotational energy; treat
  black holes as terminal reserves, not renewable ones.
- Dark: operate at the de Sitter horizon's floor; accept this as
  survival at the edge of the possible, not business as usual.

## Hard Constraints
-  $H(a) = 0$  if action  $a$  assumes infinite energy, infinite
  computation, or literal eternity (Eq. 8).
-  $H(a) = 0$  if action  $a$  cannot be distinguished, even in principle,
  from a thermal (Boltzmann) fluctuation rather than a genuine
  continuer of prior identity/information (Sec. 6.4).

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6 Escape Hatches, and Why We Flag Them as Speculative

A responsible treatment of this question has to mention, and clearly label, the handful of proposals in the literature that claim to circumvent Section 4's ceiling entirely—while being explicit that none currently commands anything resembling consensus support.

6.1 Tipler's Omega Point

Tipler [19] proposed that a universe which eventually recollapses in a “Big Crunch” could, in its final moments, support an unboundedly large, indeed formally infinite, amount of computation as the collapse accelerated toward a final singularity—an argument structurally similar to Dyson's, but run in the opposite temporal direction (toward a hot, dense end state rather than a cold, dilute one). The proposal requires a

closed, recollapsing universe, which is directly inconsistent with the observed accelerating expansion that motivates Section 4, and it has additionally been criticized on independent physical and logical grounds by other physicists and philosophers of physics, including detailed critiques of the thermodynamic and cosmological assumptions underlying the argument.

6.2 Cyclic Cosmology

Penrose’s Conformal Cyclic Cosmology [16] proposes that the remote, maximally-diluted future of one cosmic aeon becomes, via a conformal rescaling that discards the notion of a fixed scale of length and time, the initial conditions—the “big bang”—of a subsequent aeon. This is an actively debated proposal within cosmology, with claimed (and disputed) observational signatures in the cosmic microwave background; it is not something a continuity program should treat as a load-bearing assumption.

6.3 Baby Universe Creation

More speculative still, proposals associated with Smolin’s “cosmological natural selection” and related work suggest that sufficiently advanced engineering (or, in Smolin’s original formulation, black-hole formation itself) might seed causally disconnected new universes, effectively allowing a civilization to “start over” with a fresh entropy budget elsewhere, unreachable and unobservable from the parent universe. This sits closer to cosmological speculation than to an engineering proposal in the sense used elsewhere in this paper, and we note it only for completeness.

6.4 A Failure Mode Worth Naming, Not Engineering Around

None of the proposals above change this paper’s core conclusion, and they are included specifically because a survey claiming to map “how far engineering can push this” would be incomplete—arguably dishonest by omission—if it did not name the proposals that claim to remove the ceiling, alongside the reasons mainstream physics treats each with caution.

There is also a failure mode internal to the Dark Era (Section 3.4) that is better named than engineered around. As entropy rises toward its maximum, ordinary thermal fluctuations become, in the strict statistical-mechanical sense, more likely to produce a momentary, spontaneous, self-aware configuration—a so-called Boltzmann brain—than any engineered lineage is likely to survive as a genuine continuer of itself over the same timescale. A continuity program judged only by whether *something* self-aware persists, with no further criterion, is not obviously distinguishable from a program that has already failed and is merely being kept company by statistical noise. This is the reason COSMOS.md above includes a second hard constraint distinguishing genuine continuation from indistinguishable thermal fluctuation—a distinction the original device-scale architecture’s goal of “preserve core identity” gestures toward but does not resolve, and this paper does not fully resolve it either; it is flagged as an open problem rather than solved.

7 Society as a Distributed, Slow, Already-Running Experiment

7.1 Humans as Parallel, Asynchronous, Already-Deployed Agents

Every disciplinary tool used in Sections 2–6—era-transition triggers, resource budgets, hard constraints—was built to govern a single engineered agent or a single coordinated civilization treated as one decision-maker. Real human societies do not work this way, and it is worth being precise about how they differ, because the difference is the paper’s opening for a compositional theory of the loop rather than only a durational one.

A human life is, functionally, a single long-running process with no checkpoint-and-restore capability: it begins, accumulates state (memory, skill, relationships, institutional position) for on the order of 10^9 seconds, and terminates, at which point almost all of that accumulated state is lost except for whatever fraction was successfully *externalized*—written down, taught to another instance, embedded in a tool, encoded in an institution. Individually, this is a strikingly bad design for a long-horizon computing substrate: no rewind, no fork, a serial reasoning speed several orders of magnitude slower than electronic computation, and a near-total loss of internal state at every termination event.

And yet the aggregate of roughly 10^{11} such lives lived to date, running with only loose, asynchronous coordination (language, trade, law, and more recently digital networks), has produced mathematics, empirical science, industrial civilization, and the very engineering literature surveyed in Sections 3–3.3. The paper’s claim in this section is that this is not a paradox to be explained away but the central design fact to be understood: **the comparative advantage of the human ensemble was never per-instance speed; it was parallelism, redundancy, and externalized checkpointing at the level of the tools and institutions built along the way—tools and institutions that, crucially, had to be built anyway, as a side effect of running the ensemble at all, and are therefore available at zero marginal cost to any subsequent effort that wants to extend or accelerate the same process.** An AI reasoning system built today does not start from a blank substrate; it starts already standing on 10^5 years of externalized human checkpoints—language, mathematics, recorded history, codified law, engineered materials—that no purely synthetic process had to derive from scratch. This is the sense in which the original loop paper’s “the tooling had to be built anyway” is not an incidental remark but the load-bearing justification for treating human society as an asset to extend, rather than a legacy system to route around.

7.2 Ashby’s Law of Requisite Variety, Applied to a Society of Slow Agents

Cybernetics supplies a precise vocabulary for why parallel, slow, imperfectly-coordinated agents can outperform a single fast one on certain classes of problem. Ashby’s law of requisite variety states, informally, that a regulating system can only successfully control a system whose variety (the number of distinguishable states it can be in) is at least as large as the variety of the disturbances it must counter [4]. A single reasoner, however fast, has a bounded variety of internal states it can explore per unit of wall-clock time, constrained by its own architecture and its own history of training or experience. A population of N weakly-coupled agents, each exploring a different region of the same state space, generates variety roughly proportional to N times the variety of any one agent, discounted by the overlap between their explorations—and human societies, precisely because coordination between individuals is loose, imperfect, and geographically and culturally fragmented, tend to have low overlap and correspondingly high aggregate variety. This is a cybernetic restatement of the everyday observation that a diverse society tries more distinct approaches to a given problem than any single, however capable, individual would think to try alone, and it is the same argument, formalized, behind the intuition that a multitude of differently-initialized, differently-perturbed agent runtimes will out-cover a single well-tuned one on an open-ended problem, a point taken up directly in Section 8.

7.3 Social Memory as a Lossy, Redundant Checkpoint Mechanism

Niklas Luhmann’s systems-theoretic account of society [12] treats social institutions—law, science, the economy, mass media—not as collections of people but as autopoietic communication systems that reproduce their own elements (further communications) over time, in a manner that survives the replacement of any individual participant. This is precisely a checkpointing mechanism in the sense used later in this paper, with three properties worth making explicit because they recur, transformed, in the computing architecture of Section 8:

- **Lossy.** Only a small fraction of any individual life’s accumulated state is successfully externalized into a form the next instance can load: most of what a person learns dies with them, and what survives (a paper, a taught skill, a law, a proverb) is a heavily compressed, lossy summary rather than a full-state dump.
- **Redundant.** The same lesson is typically re-encoded independently by many individuals and institutions (a piece of practical wisdom appears in many cultures’ proverbs; a scientific finding is often independently rediscovered), which is wasteful of individual effort but robust to the loss of any single carrier—a property directly analogous to redundant checkpointing across unreliable nodes in distributed computing.
- **Asynchronous and version-skewed.** Different individuals and institutions load different, partially out-dated snapshots of the social checkpoint at different times, which is a source of the very failure mode the prompt for this section names directly: because no two people ever load an identical, fully up-to-date state, whole populations can and routinely do “hit the same stone twice”—rediscovering known failure modes (financial bubbles, certain categories of political failure, certain personal or organizational mistakes) that were, in principle, already checkpointed as lessons somewhere in the social memory, but not successfully loaded by the instance that needed them.

7.4 Why Repetition Is Not Purely a Defect

It would be easy to treat the previous paragraph’s third property as an unambiguous inefficiency to be engineered away, and Section 8 does propose mechanisms to reduce it. But repetition of a known failure by a new instance is not purely wasted computation, for two reasons that carry directly into the multi-agent architecture proposed below. First, a lesson re-derived from direct experience is, empirically, retained and propagated differently than one only received second-hand—a point with support in the organizational-learning literature on the difference between codified and experiential knowledge [2]. Second, and more directly relevant to this paper’s framing, a population in which every instance perfectly loaded every prior lesson before acting would be a population with reduced variety in Ashby’s sense: it would converge quickly onto a single dominant strategy and lose the exploratory redundancy that Section 7.2 identified as the actual source of the ensemble’s comparative advantage. A society that never repeats a mistake is not obviously the same thing as a society that maximizes long-run coverage of the space of possible strategies; the tension between these two goals—minimize wasted repetition versus preserve exploratory variety—is exactly the tension a multi-agent recreation architecture has to manage explicitly, rather than resolve by simply eliminating repetition, and it is the reason the checkpoint-and-rewind mechanism proposed in Section 8 is designed to *dampen*, not eliminate, the recurrence of known failure modes.

8 Recreating Human Experience Across Agent Runtimes: Checkpoints, Rewinds, Forks

8.1 From Metaphor to Architecture

Section 7 argued that human society already implements a distributed, checkpointed, imperfectly-coordinated multi-agent process, and that this process’s comparative advantage lies in coverage rather than speed. This section takes the metaphor literally and specifies a computing architecture, compatible with the Outer Loop / Inner Loop / bias-field formalism of Section 5, for deliberately recreating and extending human experience across a multitude of agent runtimes—not replacing the biological process, but running additional, synthetic instances of it, at a speed and with a rewind capability the biological process itself does not have.

Definition 1 (Runtime). A runtime r is a single instantiation of an agent—biological or synthetic—together with its accumulated internal state $\sigma(t)$ at time t , evolving under a policy π_r and an environment $\mathcal{E}_r$. A human life is a runtime with π_r realized biologically and $\mathcal{E}_r$ realized as the physical and social world; an AI agent instance is a runtime with π_r realized computationally.

Definition 2 (Checkpoint). A checkpoint $c = (\sigma(t), t, \mathcal{E}_r(t))$ is a serialized snapshot of a runtime’s internal state and environment at time t , sufficient to resume or fork execution from that point. For a biological runtime, only a lossy projection $\hat{\sigma}(t) \ll \sigma(t)$ is checkpoint-able (Section 7.3); for a synthetic runtime built on the architecture proposed here, $\sigma(t)$ can in principle be checkpointed in full.

Definition 3 (Fork). A fork of runtime r at checkpoint c is a new runtime r' initialized from c but permitted to diverge under an altered policy $\pi_{r'} \neq \pi_r$, an altered environment $\mathcal{E}_{r'} \neq \mathcal{E}_r$, or both. The fork shares r ’s full history up to t and no history after t .

Definition 4 (Rewind). A rewind of runtime r to checkpoint c at time $t' > t$ discards the trajectory $\sigma(t \rightarrow t')$ and resumes execution from c , typically triggered by a detector recognizing that the discarded trajectory is repeating a previously checkpointed failure mode (Section 7.4).

8.2 Why Recreate Human Experience Specifically, Rather Than Optimize a Synthetic Policy Directly

A natural objection to this architecture is that if the goal is broad exploration of a state space, a synthetic multi-agent system could simply be designed and optimized directly, without the detour of recreating human experience as such. Two considerations from Sections 7 and 5 argue against this shortcut. First, the “tooling had to be built anyway” argument of Section 7.1: human experience is not merely one dataset among others, it is the dataset embedding 10^5 years of externalized, socially-validated checkpoints—language, causal reasoning about the physical and social world, ethical and legal precedent—at zero marginal construction cost to any process that can load it, whereas a synthetic policy optimized from scratch has to rediscover an unknown but plausibly very large fraction of that content de novo, paying an energy and time cost against the very budget constrained in Section 4. Second, the diversity argument of Section 7.2: a directly-optimized synthetic system, absent an explicit and difficult-to-specify diversity objective, tends to converge on a narrower region of policy space than a population initialized from many distinct, already-diverse human trajectories, precisely because those trajectories encode variety accumulated through genuinely independent lived experience rather than variety imposed after the fact by an optimizer’s regularization term.

8.3 A Reference Architecture

We propose a reference architecture, summarized in Table 2, mapping each concept above onto the Outer Loop / Inner Loop distinction of Section 5.

8.4 Rewinds and the “Same Stone Twice” Problem

Section 7.4 argued that repetition of known failure modes is not purely wasteful, because eliminating it entirely would reduce exploratory variety. The architecture proposed here therefore does not attempt to prevent forks from ever repeating a mistake; it attempts to *bound how much budget any single repetition is allowed to consume* before a rewind is triggered. Concretely, we define a divergence score

$$D(r, t) = \max_{c_i \in \mathcal{C}_{\text{failure}}} \text{sim}(\sigma_r(t), c_i) \quad (7)$$

Layer	Function	Relation to Section 5
Runtime pool manager	Instantiates new forks from a library of checkpoints; allocates compute budget across concurrently running forks	Outer Loop: analogue of the goal-priority queue, here allocating exploration budget instead of energy
Checkpoint library	Stores $(\sigma, t, \mathcal{E})$ triples with provenance metadata (which prior forks and, where available, which human experiential data contributed to $\hat{\sigma}$)	Corresponds to the “resource state” tracked by the Outer Loop
Divergence detector	Monitors running forks for repetition of a previously logged failure signature; triggers a rewind (Section 8.4) rather than allowing indefinite repetition	Inner Loop’s Failure Detection & Reboot Logic (Section 3.1 of the original architecture), generalized from hardware failure to strategy failure
Curiosity-weighted scheduler	Allocates the next unit of exploration budget preferentially toward forks bordering unexplored, high-information-value regions of the state space (formalized in Section 9)	Extends the bias field (Eq. 5) with an explicit information-value term
Provenance / identity ledger	Tracks the fork lineage of every runtime, so that downstream analysis (and the philosophical treatment of Section 9.3) can distinguish a genuine continuer of a given trajectory from an unrelated runtime that merely converged to a similar state	Novel to this layer; has no direct analogue in the original device- or cosmological-scale architecture

Table 2: Reference architecture for recreating human experience across a multitude of agent runtimes.

where $\mathcal{C}_{\text{failure}}$ is the library of checkpoints previously tagged, by any prior runtime’s outcome, as leading to a logged failure mode, and $\text{sim}(\cdot, \cdot)$ is a similarity measure over runtime states. A rewind is triggered when $D(r, t)$ exceeds a threshold *and* the runtime’s remaining exploration budget (Section 8.5) is judged better spent on a fresh fork than on continuing to observe a highly likely repetition play out in full. Crucially, the threshold is not zero: a runtime is permitted to venture some distance into previously-failed territory, both because similarity is an imperfect predictor of actual outcome (the same surface conditions do not guarantee the same result) and because, per Section 7.4, letting some fraction of forks rediscover a lesson directly, rather than only inheriting it as a checkpoint, has its own value. Equation 7 is therefore best read as an explicit, tunable answer to a question human societies have never been able to set deliberately: exactly how much of the “same stone” should any given instance be allowed to hit before the system intervenes.

8.5 Budgeting Exploration Against the Ceiling of Section 4

The runtime pool manager’s allocation problem is not free: every fork consumes compute, and compute consumes energy, and Section 4 established that total available energy—and, at the largest scale, total available entropy—is finite even in the most optimistic engineering program this paper considers. The exploration-budget allocation problem facing the runtime pool manager in, say, the year 2100 is a small-scale, near-term instance of the same allocation problem facing the Outer Loop of Section 5 across cosmological eras: both are asking “how much of a finite, non-renewable resource should be spent generating variety now, versus preserved for a later point at which more may be known about which variety is worth having.” We treat this formally in Section 10.3.

8.6 RUNTIME.md: A Rules File for the Recreation Architecture

Following the paper's established practice of writing each layer's governing rules down as an explicit, auditable artifact rather than leaving them implicit in code, we specify a RUNTIME.md for this layer, parallel in spirit to LOOP.md and COSMOS.md:

```
# RUNTIME.md -- Multi-Agent Human-Experience Recreation Rules

## Core Principle
Value lives in the ensemble of forks, not in any single trajectory.
Do not optimize away redundancy or repetition without first checking
whether it is preserving exploratory variety (Sec. 6.2 Ashby argument).

## Checkpoint Library
- Every runtime state that is externalized (a lesson, a tool, a
  written record) is a checkpoint candidate.
- Tag checkpoints with provenance: which prior forks, and which
  human-experiential data, contributed.
- Tag failure checkpoints (C_failure) separately; these feed the
  divergence detector, not the general library.

## Fork Policy
- New forks may diverge in policy, environment, or both.
- A fork inherits full history up to its checkpoint and no history
  after it (Def. 3). Do not silently merge post-fork histories.

## Divergence Detector / Rewind Policy
- Compute  $D(r,t) = \max$  similarity to any C_failure checkpoint (Eq. 9).
- Trigger rewind only if  $D(r,t)$  exceeds threshold AND remaining
  budget favors a fresh fork over letting the trajectory play out.
- Threshold is non-zero by design: some direct rediscovery of known
  failure modes is retained, not eliminated (Sec. 5.4 repetition
  argument).

## Curiosity-Weighted Scheduling
- Allocate exploration budget using  $\kappa(a)$ , expected uncertainty
  reduction (Eq. 10), not extrinsic reward alone.
- Treat cross-fork convergence on the same underexplored region as
  a new edge in the provenance/exploration network (Sec. 7.2).

## Hard Constraints
- Do not assume unlimited exploration budget; weight  $\kappa(a)$ 
  against the same  $\lambda(e)$  discount used in the cosmological
  bias field (Eq. 6).
- Do not silently resolve fork-identity questions (Sec. 7.3); make
  any continuer policy explicit and auditable in the provenance
  ledger, not implicit in scheduler behavior.
```

9 Curiosity as the Mechanism That Builds a Network, Not Only a Drive

9.1 Curiosity as Value of Information

The preceding two sections established a population of forking, checkpointing, occasionally-rewound run-times. Left unaddressed is what determines *where* a fork explores next—the curiosity-weighted scheduler referenced in Table 2 but not yet formalized. We adopt the value-of-information framing common to Bayesian experimental design and to computational models of curiosity in reinforcement learning [14, 17]: an action or exploratory step is assigned intrinsic value in proportion to the reduction in uncertainty it is expected to produce about the state space, independent of any extrinsic reward. Formally, if $\mathcal{U}(s)$ denotes the scheduler’s uncertainty over the outcome distribution at state s , the curiosity-weighted term added to the bias field of Equation 5 is

$$\kappa(a) = \mathbb{E}[\mathcal{U}(s) - \mathcal{U}(s') \mid a] \quad (8)$$

the expected reduction in uncertainty from taking action a and transitioning from state s to s' . A scheduler that maximizes only extrinsic task reward will, absent this term, tend to reproduce the low-variety convergence problem identified in Section 7.2; a scheduler that weights $\kappa(a)$ non-trivially against extrinsic reward is, in effect, deliberately re-injecting the exploratory redundancy that human society generates for free through the mere fact of having many independent, only loosely coordinated instances.

9.2 Curiosity Builds Networks, Not Just Discoveries

The prompt motivating this section observes that curiosity does not only drive individual discovery, it builds *networks*—connections between otherwise separate lines of inquiry, and between otherwise separate people. This is well supported in the sociology of science: Diana Crane’s study of “invisible colleges” [6] documents how curiosity-driven contact across otherwise disconnected research groups is the primary mechanism by which isolated findings become a cumulative, cross-referencing body of knowledge, rather than a pile of unconnected results; more recent network-scientific treatments of collaboration graphs make the same point quantitatively; curious, information-seeking behavior correlates strongly with the formation of new ties in a collaboration network, more strongly than institutional proximity alone [13]. Applied to the architecture of Section 8, this suggests that the curiosity-weighted scheduler of Equation 8 is doing double duty: it is not only steering individual forks toward high-information-value states, it is, whenever two forks are drawn toward the same underexplored region from different starting checkpoints, creating exactly the kind of cross-fork contact that the provenance ledger of Table 2 can register as a new edge in a growing exploration network—turning what would otherwise be a forest of disconnected trajectories into a connected graph, which is a precondition for the redundant-but-cross-referencing checkpointing behavior described sociologically in Section 7.3.

9.3 The Philosophical Problem: Who Is the Continuer?

A fork, as defined in Section 8.1, shares a full history up to the checkpoint and no history after it. This is precisely the structure of the thought experiments Derek Parfit used to argue that personal identity is not the deep further fact ordinary intuition takes it to be, but rather a matter of degree, constituted by overlapping chains of psychological continuity and connectedness [15]. Parfit’s own preferred conclusion—that that Relation R (psychological continuity and connectedness, whatever its cause) is “what matters” in survival, and that the question “but is it really me” can be a question without a determinate answer—was developed for a case of exactly one branching event. The architecture of Section 8 generalizes this to a population in continuous, ongoing forking, and this generalization surfaces a difficulty Parfit’s framework does not, by

itself, resolve: if value is to be assigned to the *ensemble* of forks (as the coverage argument of Section 7.2 recommends) rather than to any privileged single trajectory, then the ordinary language of “the agent’s welfare” or “the agent’s goals persisting” stops picking out a well-defined referent, and has to be replaced with a claim about the welfare or goal-satisfaction of a *distribution* over continuers. We do not resolve this here; we flag it, in the same spirit as the Boltzmann-brain problem of Section 6.4, as a genuine open problem that any real deployment of the architecture in Section 8 would have to take a position on, most likely by extending the provenance ledger into an explicit, auditable policy about which forks count as continuers of a given prior life or agent for whatever ethical or legal purposes matter to the deployers—a policy question, not merely an engineering one.

9.4 Curiosity Under a Finite Budget

Section 8.5 noted that exploration budget is finite. Equation 8’s uncertainty-reduction term therefore cannot be maximized without bound; it must itself be weighed, in the full bias field, against the same $\lambda(e)$ discount introduced in Equation 5. This produces a clean statement of the tension this section has been building toward: curiosity is what makes the ensemble of runtimes worth more than the sum of its parts (Section 9.2), but curiosity, like every other term in the bias field, is a claim on a finite resource, and a civilization—or a runtime pool manager, at any scale from a research lab to the cosmological Outer Loop of Section 5—has to decide, explicitly and non-arbitrarily, how much of that resource curiosity is entitled to at each era of the loop’s operation.

9.5 Population Diversity as a Quantity Distinct from Curiosity

Equation 8 defines $\kappa(a)$ at the level of a single runtime’s expected uncertainty reduction. It is worth being precise that this is not the same quantity as the population-level diversity invoked informally in Section 7.2 and flagged as a distinct Outer Loop trigger in Section 5.2, because the two can diverge: a population in which every fork is individually pursuing high- κ actions can nonetheless converge on a single dominant strategy if every fork happens to resolve its uncertainty in the same direction, at which point the population has lost the coverage advantage of Section 7.2 even though no single runtime’s scheduler is behaving myopically. We define a population-level diversity index over the set of concurrently active runtimes $\mathcal{R}(t) = \{r_1, \dots, r_N\}$ at time t :

$$V(t) = - \sum_k p_k(t) \log p_k(t), \quad p_k(t) = \frac{|\{r \in \mathcal{R}(t) : \sigma_r(t) \in \mathcal{B}_k\}|}{N} \quad (9)$$

an entropy over a partition of runtime-state space into bins $\{\mathcal{B}_k\}$, so that $V(t)$ is maximal when active runtimes are spread evenly across distinct regions of the state space and falls toward zero as they converge on a shared region. $V(t)$ is a property of the *population*, computable from the checkpoint library’s provenance metadata (Table 2) without reference to any individual runtime’s internal uncertainty estimate, and it is $V(t)$, not $\kappa(a)$, that the “diversity collapse” trigger of Section 5.2 monitors: a floor condition $V(t) < V_{\min}$ is what should prompt the Outer Loop to intervene—for instance by explicitly forking new runtimes from under-represented regions of the checkpoint library, rather than simply trusting the existing population’s own curiosity-weighted schedulers to correct the collapse from within, since Equation 8 alone, as just argued, does not guarantee $V(t)$ stays high. This is the sense in which diversity maintenance is a genuine addition to the bias field rather than a restatement of curiosity under another name: $\kappa(a)$ governs what a single fork does next, $V(t)$ governs whether the ensemble of forks, taken together, is still worth running as a population at all.

Section 10.3 returns to $\lambda(e)$, $\kappa(a)$, and $V(t)$ together as the paper’s final integration point.

10 Discussion

10.1 The Limiting Factor Has Moved

The exercise this paper set out to perform—push the original loop architecture as far as real physics allows—lands in an unusual place for what began as an engineering paper. The limiting factor, more than seventy orders of magnitude out from the present, is not manufacturing capability, materials science, or even raw energy-collection technology. Each of those is, in principle, a solvable problem given sufficient time and a sufficiently large civilization, and Sections 3.1–3.3 describe how. The limiting factor identified in Section 4 is instead a property of the geometry of spacetime itself under a positive cosmological constant. This is a stronger and stranger version of the original paper’s conclusion, not a contradiction of it: the original paper argued a drone cannot beat the Second Law over the course of a few days; this paper argues that a maximally sophisticated, multi-galactic civilization cannot beat the finite entropy bound of de Sitter space over 10^{100} years, for a structurally similar underlying reason—the constraint is on the container the loop runs inside, not on the sophistication of the loop itself.

10.2 Why “Finite” Is Not a Downgrade

“Finite, but astronomically large” is an unusual thing to characterize as a limitation. The ratio between a human lifespan and the timescale of the de Sitter horizon’s relevant physics is larger than the ratio between a single heartbeat and the entire age of the universe elapsed so far, repeated over and over. A civilization that actually executed the program of Section 3 would, from any human vantage point, be functionally indistinguishable from eternal. The insistence in Section 4 on the word *finite* is not pessimism; it is the same discipline the original device-scale paper applied when it declined to round its own conclusion up to “impossible” or down to a false “solved.” Stating the true boundary precisely, at whatever scale one is operating, is the more useful output than stating a comforting approximation of it.

10.3 What Carries Over from Device Scale to Cosmological Scale

Three structural features of the original architecture turn out to generalize essentially unchanged: (i) the two-tier separation of long-horizon goal management from short-horizon task execution: unchanged in structure across seventy orders of magnitude of rescaling; (ii) the probabilistic bias field as a way of blending hard constraints with soft, cost-sensitive preferences, extended only by an explicit budget-discount term (Equation 5); and (iii) the practice of writing the architecture’s rules down as an explicit, auditable artifact (`LOOP.md` at device scale, `COSMOS.md` here), rather than leaving it implicit in code or in an agent’s behavior. What does *not* carry over unchanged is the content of the hard constraints themselves: at device scale, safety constraints are about not harming the people and environment around the device; at cosmological scale, the load-bearing hard constraint (Equation 6) is instead an epistemic one, about never planning as though a false cosmological premise were true.

10.4 The Durational and Compositional Halves, Integrated

Sections 2–6 and Sections 7–9 were developed with different vocabularies—cosmology and control theory in the first half, sociology, cybernetics, computing, and philosophy in the second—but they are answers to the same underlying allocation problem, restated at different scales, and it is worth making the correspondence explicit rather than leaving the two halves of the paper as a durational analysis bolted onto a compositional one.

- The Outer Loop’s era-transition re-planning (Section 5.1) and the runtime pool manager’s budget allocation across forks (Section 8.5) are the same operation—deciding how much of a finite resource to spend now versus preserve—performed at timescales differing by roughly seventy orders of magnitude.
- The bias field’s hard constraint against planning on infinite energy (Equation 6) and the curiosity-weighted scheduler’s own budget constraint (Section 9.4) are the same constraint: no layer of the architecture, from a single research program to a Dark Era civilization, is exempt from planning against a genuinely finite budget.
- The Boltzmann-brain problem (Section 6.4)—distinguishing a genuine continuer from an indistinguishable thermal fluctuation—and the fork-identity problem (Section 9.3)—distinguishing a genuine continuer of a life or agent from an unrelated runtime that merely converged to a similar state—are, formally, the same open problem, encountered once at the scale of a single provenance ledger and once at the scale of the entropy of the universe. Neither this paper nor, to our knowledge, the literature it draws on fully resolves it; both instances are flagged rather than solved, deliberately, because a false resolution would be worse than an acknowledged gap.
- Ashby’s variety argument (Section 7.2), applied to human society, and the “expand early to maximize reachable resources” argument (Proposition 1), applied to a Stelliferous-era civilization, share a common structure: both argue that a strategy which looks locally inefficient—tolerating redundant, uncoordinated, loosely-synchronized parallel effort in one case; spending heavily on expansion before local optimization in the other—dominates a strategy that looks locally efficient, once the relevant resource (variety of explored states; reachable galaxies before horizon recession) is understood to be forfeited irreversibly if not claimed early.
- The sociological triggers of Section 5.2 and the population-diversity index $V(t)$ of Section 9.5 are, together, the compositional-half analogue of the physical era-transition triggers of Section 5.1: just as the Outer Loop must monitor the physical substrate for state changes it cannot itself reverse (a star exhausting its fuel), it must independently monitor the population of runtimes for state changes that leave the physical budget untouched but empty out the reason the budget was worth spending in the first place. A civilization that tracks $\lambda(e)$ and Equation 3 carefully while letting $V(t)$ collapse unnoticed has solved the durational problem of Sections 2–6 while quietly failing the compositional one of Sections 7–9.

The upshot is that this paper’s two halves are not, in the end, two different papers awkwardly combined: they are the same finite-resource allocation problem, examined once at the scale of a civilization harvesting Hawking radiation in the year 10^{80} , and once at the scale of a research group, an institution, or a population of forked agent runtimes deciding, this year, how much curiosity and diversity it can afford to fund out of the same finite budget that also has to pay for everything else the loop needs to survive.

11 Conclusion

This paper asked two questions that turned out to be one question asked twice, at radically different scales. The first, durational question was how far a self-sustaining loop architecture could be pushed if an entire civilization’s engineering capacity were devoted to it across the full remaining lifetime of the universe. Using currently mainstream cosmology, the answer runs through four increasingly strange energy regimes—Dyson swarms and stellar engines in the Stelliferous era, reversible slowing computation in the Degenerate era, Hawking-radiation harvesting in the Black Hole era—and arrives, in the Dark Era, at a hard, geometry-imposed ceiling on total information processing, set by the finite entropy of the de Sitter horizon of an

accelerating universe (Proposition 2). The loop can, on this analysis, be extended by a factor on the order of 10^{98} relative to a human lifespan, but not past that ceiling into literal infinity.

The second, compositional question was what such a loop should be made of, and here the paper’s answer is that humanity has already been running the relevant architecture, imperfectly and without deliberate design, for 10^5 years: a massively parallel population of slow, non-restartable, lossily-checkpointing runtimes whose aggregate output has repeatedly outstripped what any single fast reasoner working alone would have produced over the same span, precisely because of redundancy and variety that a more centrally optimized process would have engineered away. Sections 7–9 proposed making this architecture deliberate rather than incidental: a reference design for forking, checkpointing, and selectively rewinding a multitude of agent runtimes recreating and extending human experience, governed by a curiosity-weighted scheduler that treats variety-seeking not as an indulgence but as the mechanism that turns an otherwise disconnected forest of trajectories into a single, cross-referencing exploration network—while confronting, rather than dissolving, the philosophical difficulty of what a “continuer” even means once forking is continuous and ongoing rather than a one-time thought experiment.

Both halves converge on the same discipline. A civilization that fails to encode Equation 6 risks spending an irrecoverable early-era surplus on the false premise that later cosmological eras will be equally generous; a research program, an institution, or a population of forked runtimes that fails to budget its curiosity (Section 9.4) against a genuinely finite exploration budget is making the identical mistake at a scale of years rather than aeons. The honest answer this paper gives, twice, at two scales seventy orders of magnitude apart, is the same one: an architecture built to last, and to be worth something across however long it lasts, has to be built around the true, finite shape of its resources—energy, entropy, attention, or curiosity—not around a more comfortable, unbounded version of any of them.

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Final note. Nothing in this paper is a plan anyone is executing. It is a survey of where physicists have taken the “how long can intelligence outlast the universe” question, assembled into the loop-architecture language of the paper it responds to. The honest ending—a finite, unimaginably large number, not infinity—is a more interesting and more useful answer than either extreme.